

# eda\_health\_fitness\_dataset\_v1.1

November 18, 2025

## 1 EDA – FitLife Health & Fitness Tracking Dataset

Ce notebook explore le dataset **FitLife Health & Fitness Tracking** dans le cadre du projet TrAIn.me.

L'objectif est de vérifier la qualité des données (structure, typage, complétude) avant toute modélisation prédictive.

### 1.1 Objectif

Vérifier que le fichier `health_fitness_dataset.csv` est correctement chargé et obtenir une vue d'ensemble :

- dimensions du jeu de données (nombre de lignes et de colonnes) ;
- typage des variables (numériques, catégorielles) ;
- première inspection des valeurs manquantes et des doublons ;
- cohérence générale du corpus avant analyse statistique détaillée.

### 1.2 Interprétation attendue

À partir de cette première inspection, il s'agira de commenter :

- la taille globale du dataset (nombre de lignes / colonnes) ;
- la nature des variables (mélange de mesures numériques, indicateurs de santé et catégories) ;
- la présence éventuelle de valeurs manquantes et leur concentration sur certaines colonnes ;
- l'existence (ou non) de doublons dans les données brutes ;
- la cohérence générale des premières lignes (valeurs réalistes, formats attendus).

Cette interprétation sert de point de départ pour juger de la qualité du corpus et décider des premières actions de nettoyage à mener dans les sections suivantes.

### 1.3 Configuration notebook

#### 1.3.1 Importation des librairies essentielles

```
[4]: import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import kagglehub
```

```

from pathlib import Path
import sys

# === Localiser automatiquement le package 'themes' en remontant l'arborescence
↳===
root = Path.cwd() # ex : .../train.me/src/notebooks/eda/health_fitness_dataset
themes_root = None

for p in [root, *root.parents]:
    if (p / "themes").is_dir():
        themes_root = p
        break

if themes_root is None:
    raise FileNotFoundError(
        f"Impossible de trouver le dossier 'themes' en partant de {root}. "
        "Vérifie l'arborescence : il doit exister un dossier 'themes/' quelque
↳part au-dessus."
    )

if str(themes_root) not in sys.path:
    sys.path.insert(0, str(themes_root))

print(" Ajouté au PYTHONPATH :", themes_root)
print(" Contenu de", themes_root, ":", [x.name for x in themes_root.iterdir()])

from themes.theme_train_me import set_trainme_theme

```

Ajouté au PYTHONPATH :

c:\Users\fbac\Desktop\Projets\Dev\GitHub\train.me\src\notebooks

Contenu de c:\Users\fbac\Desktop\Projets\Dev\GitHub\train.me\src\notebooks :

['eda', 'model\_training', 'preprocessing', 'themes', '\_\_init\_\_.py']

c:\Users\fbac\Desktop\Projets\Dev\GitHub\train.me\conda\Lib\site-packages\tqdm\auto.py:21: TqdmWarning: IProgress not found. Please update jupyter and ipywidgets. See [https://ipywidgets.readthedocs.io/en/stable/user\\_install.html](https://ipywidgets.readthedocs.io/en/stable/user_install.html)

from .autonotebook import tqdm as notebook\_tqdm

### Configuration d'affichage et thème TrAIIn.me

```

[5]: # Options d'affichage pour les tableaux
pd.set_option("display.max_columns", None)
pd.set_option("display.width", 120)

# =====
# Thème TrAIIn.me (Seaborn/Matplotlib)
# =====

```

```
USE_DARK_THEME = True # bascule possible vers un thème clair si besoin

palette_trainme = set_trainme_theme(
    context="talk",
    font_scale=1.05,
    use_dark=USE_DARK_THEME
)

palette_trainme
```

[5]: ['#00E5FF', '#00BFFF', '#0A84FF', '#1E90FF', '#00FFFF', '#64D8FF']

### 1.3.2 Activation du thème TrAIIn.me

```
[12]: if USE_DARK_THEME:
    palette_trainme = set_trainme_theme(context="talk", font_scale=1.05,
    ↪ use_dark=True)
else:
    sns.set_style("ticks") # style clair sobre, axes renforcés
    sns.set_context("talk", font_scale=1.05)
    palette_trainme = sns.color_palette("deep") # palette par défaut sobre
    print(" Style activé : 'ticks' → sobre, axes renforcés (présentation pro)")

# Exemple de vérification :
print("Palette active :", palette_trainme.as_hex() if hasattr(palette_trainme,
    ↪ "as_hex") else palette_trainme)
```

Palette active : ['#00E5FF', '#00BFFF', '#0A84FF', '#1E90FF', '#00FFFF', '#64D8FF']

## 1.4 Chargement et inspection initiale

### Chargement du dataset health\_fitness\_dataset.csv

```
[6]: # Téléchargement du dataset depuis Kaggle
dataset_path = kagglehub.dataset_download("jijagallery/
    ↪ fitlife-health-and-fitness-tracking-dataset")

# Recherche du fichier CSV dans le dossier téléchargé
csv_files = [f for f in os.listdir(dataset_path) if f.endswith(".csv")]
print("Fichiers trouvés :", csv_files)

# Construction du chemin complet vers le CSV
file_path = os.path.join(dataset_path, csv_files[0])

# Lecture du fichier CSV
df = pd.read_csv(file_path)
```

```
# Informations de confirmation
print("Dataset chargé avec succès !")
print(f"Dimensions : {df.shape[0]} lignes et {df.shape[1]} colonnes\n")
```

Downloading from  
[https://www.kaggle.com/api/v1/datasets/download/jijagallery/fitlife-health-and-fitness-tracking-dataset?dataset\\_version\\_number=1...](https://www.kaggle.com/api/v1/datasets/download/jijagallery/fitlife-health-and-fitness-tracking-dataset?dataset_version_number=1...)

100%| | 15.4M/15.4M [00:03<00:00, 5.10MB/s]

Extracting files...

Fichiers trouvés : ['health\_fitness\_dataset.csv']

Dataset chargé avec succès !

Dimensions : 687701 lignes et 22 colonnes

### Aperçu des premières lignes

```
[7]: df.head(10)
```

```
[7]:   participant_id      date  age gender  height_cm  weight_kg  activity_type
duration_minutes intensity \
0                1  2024-01-01   56     F    165.3    53.7      Dancing
41             Low
1                1  2024-01-04   56     F    165.3    53.9      Swimming
28             Low
2                1  2024-01-05   56     F    165.3    54.2      Swimming
21           Medium
3                1  2024-01-07   56     F    165.3    54.4  Weight Training
99           Medium
4                1  2024-01-09   56     F    165.3    54.7      Swimming
100          Medium
5                1  2024-01-10   56     F    165.3    54.9      HIIT
31           Medium
6                1  2024-01-11   56     F    165.3    55.2  Weight Training
97            High
7                1  2024-01-12   56     F    165.3    55.5      HIIT
70            Low
8                1  2024-01-17   56     F    165.3    55.7      HIIT
89           Medium
9                1  2024-01-18   56     F    165.3    56.0  Weight Training
115          Medium

      calories_burned  avg_heart_rate  hours_sleep  stress_level  daily_steps
hydration_level  bmi  resting_heart_rate \
0                3.3          103          6.6          3          7128
1.5  19.6          69.5
```

|     |      |      |      |     |    |       |
|-----|------|------|------|-----|----|-------|
| 1   |      | 2.9  | 102  | 8.1 | 7  | 7925  |
| 1.8 | 19.6 |      | 69.5 |     |    |       |
| 2   |      | 2.6  | 126  | 6.2 | 7  | 7557  |
| 2.7 | 19.6 |      | 69.5 |     |    |       |
| 3   |      | 10.7 | 141  | 7.2 | 8  | 11120 |
| 2.6 | 19.6 |      | 69.5 |     |    |       |
| 4   |      | 12.7 | 112  | 7.1 | 1  | 5406  |
| 1.5 | 19.6 |      | 69.5 |     |    |       |
| 5   |      | 6.8  | 121  | 7.5 | 10 | 10202 |
| 2.2 | 19.6 |      | 69.5 |     |    |       |
| 6   |      | 12.4 | 145  | 6.6 | 8  | 5912  |
| 2.8 | 19.6 |      | 69.5 |     |    |       |
| 7   |      | 12.9 | 99   | 6.1 | 7  | 9477  |
| 1.6 | 19.6 |      | 69.5 |     |    |       |
| 8   |      | 19.7 | 112  | 7.2 | 7  | 9710  |
| 3.3 | 19.6 |      | 69.5 |     |    |       |
| 9   |      | 12.8 | 117  | 5.6 | 3  | 7830  |
| 2.0 | 19.6 |      | 69.5 |     |    |       |

|                | blood_pressure_systolic | blood_pressure_diastolic | health_condition |
|----------------|-------------------------|--------------------------|------------------|
| smoking_status | fitness_level           |                          |                  |
| 0              | 110.7                   | 72.9                     | NaN              |
| Never          | 0.04                    |                          |                  |
| 1              | 110.7                   | 72.9                     | NaN              |
| Never          | 0.07                    |                          |                  |
| 2              | 110.7                   | 72.9                     | NaN              |
| Never          | 0.09                    |                          |                  |
| 3              | 110.7                   | 72.9                     | NaN              |
| Never          | 0.21                    |                          |                  |
| 4              | 110.7                   | 72.9                     | NaN              |
| Never          | 0.33                    |                          |                  |
| 5              | 110.7                   | 72.9                     | NaN              |
| Never          | 0.37                    |                          |                  |
| 6              | 110.7                   | 72.9                     | NaN              |
| Never          | 0.51                    |                          |                  |
| 7              | 110.7                   | 72.9                     | NaN              |
| Never          | 0.58                    |                          |                  |
| 8              | 110.7                   | 72.9                     | NaN              |
| Never          | 0.68                    |                          |                  |
| 9              | 110.7                   | 72.9                     | NaN              |
| Never          | 0.82                    |                          |                  |

### Informations sur le typage, les valeurs manquantes et les doublons

```
[8]: # Typage et complétude globale
df_info = df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 687701 entries, 0 to 687700

Data columns (total 22 columns):

| #  | Column                   | Non-Null Count  | Dtype   |
|----|--------------------------|-----------------|---------|
| 0  | participant_id           | 687701 non-null | int64   |
| 1  | date                     | 687701 non-null | object  |
| 2  | age                      | 687701 non-null | int64   |
| 3  | gender                   | 687701 non-null | object  |
| 4  | height_cm                | 687701 non-null | float64 |
| 5  | weight_kg                | 687701 non-null | float64 |
| 6  | activity_type            | 687701 non-null | object  |
| 7  | duration_minutes         | 687701 non-null | int64   |
| 8  | intensity                | 687701 non-null | object  |
| 9  | calories_burned          | 687701 non-null | float64 |
| 10 | avg_heart_rate           | 687701 non-null | int64   |
| 11 | hours_sleep              | 687701 non-null | float64 |
| 12 | stress_level             | 687701 non-null | int64   |
| 13 | daily_steps              | 687701 non-null | int64   |
| 14 | hydration_level          | 687701 non-null | float64 |
| 15 | bmi                      | 687701 non-null | float64 |
| 16 | resting_heart_rate       | 687701 non-null | float64 |
| 17 | blood_pressure_systolic  | 687701 non-null | float64 |
| 18 | blood_pressure_diastolic | 687701 non-null | float64 |
| 19 | health_condition         | 197426 non-null | object  |
| 20 | smoking_status           | 687701 non-null | object  |
| 21 | fitness_level            | 687701 non-null | float64 |

dtypes: float64(10), int64(6), object(6)

memory usage: 115.4+ MB

#### 1.4.1 Statistiques descriptives des variables numériques

```
[9]: display(df.describe().T)
```

|                  | count    | mean        | std        | min    | 25%    | 50%     | 75%     | max     |
|------------------|----------|-------------|------------|--------|--------|---------|---------|---------|
| participant_id   | 687701.0 | 1499.781828 | 865.997215 | 1.00   | 749.00 | 1499.00 | 2249.00 | 3000.00 |
| age              | 687701.0 | 41.658602   | 13.581770  | 18.00  | 30.00  | 42.00   | 53.00   | 64.00   |
| height_cm        | 687701.0 | 168.587699  | 9.140811   | 145.00 | 161.70 | 168.20  | 175.30  | 198.50  |
| weight_kg        | 687701.0 | 94.921981   | 22.461801  | 45.30  | 78.20  | 94.60   | 110.50  | 188.40  |
| duration_minutes | 687701.0 | 70.011671   | 29.147251  | 20.00  | 45.00  | 70.00   | 95.00   | 120.00  |
| calories_burned  | 687701.0 | 15.381302   | 9.985552   | 0.80   | 7.80   | 13.00   | 20.70   | 92.00   |

|                          |          |             |             |         |         |   |
|--------------------------|----------|-------------|-------------|---------|---------|---|
| avg_heart_rate           | 687701.0 | 131.454165  | 17.814744   | 82.00   | 118.00  | ␣ |
| ↪ 130.00                 | 144.00   | 206.00      |             |         |         |   |
| hours_sleep              | 687701.0 | 7.048799    | 0.972068    | 4.00    | 6.40    | ␣ |
| ↪ 7.00                   | 7.70     | 10.00       |             |         |         |   |
| stress_level             | 687701.0 | 5.252389    | 2.770029    | 1.00    | 3.00    | ␣ |
| ↪ 5.00                   | 8.00     | 10.00       |             |         |         |   |
| daily_steps              | 687701.0 | 8628.370918 | 2054.756608 | -419.00 | 7203.00 | ␣ |
| ↪ 8607.00                | 10027.00 | 17241.00    |             |         |         |   |
| hydration_level          | 687701.0 | 2.499427    | 0.579050    | 1.50    | 2.00    | ␣ |
| ↪ 2.50                   | 3.00     | 3.50        |             |         |         |   |
| bmi                      | 687701.0 | 22.733134   | 3.568383    | 14.20   | 20.10   | ␣ |
| ↪ 22.40                  | 25.10    | 38.80       |             |         |         |   |
| resting_heart_rate       | 687701.0 | 70.010547   | 5.074563    | 51.10   | 66.50   | ␣ |
| ↪ 70.00                  | 73.50    | 87.10       |             |         |         |   |
| blood_pressure_systolic  | 687701.0 | 120.001260  | 10.008917   | 78.00   | 113.10  | ␣ |
| ↪ 120.20                 | 127.00   | 152.70      |             |         |         |   |
| blood_pressure_diastolic | 687701.0 | 80.188079   | 8.239418    | 53.70   | 74.60   | ␣ |
| ↪ 80.10                  | 85.70    | 112.10      |             |         |         |   |
| fitness_level            | 687701.0 | 9.524900    | 5.502485    | 0.02    | 4.77    | ␣ |
| ↪ 9.51                   | 14.23    | 21.93       |             |         |         |   |

#### 1.4.2 Analyse des valeurs manquantes et doublons

```
[10]: # Comptage des valeurs manquantes par colonne
missing_counts = df.isna().sum().sort_values(ascending=False)

print("\n Valeurs manquantes par colonne (top 20) :")
display(missing_counts.head(20))

# Comptage des doublons ligne à ligne
n_duplicates = df.duplicated().sum()
print(f"\n Nombre de doublons (lignes strictement identiques) :␣
↪{n_duplicates}")
```

Valeurs manquantes par colonne (top 20) :

|                          |        |
|--------------------------|--------|
| health_condition         | 490275 |
| participant_id           | 0      |
| date                     | 0      |
| smoking_status           | 0      |
| blood_pressure_diastolic | 0      |
| blood_pressure_systolic  | 0      |
| resting_heart_rate       | 0      |
| bmi                      | 0      |
| hydration_level          | 0      |
| daily_steps              | 0      |
| stress_level             | 0      |

```

hours_sleep          0
avg_heart_rate       0
calories_burned      0
intensity            0
duration_minutes     0
activity_type        0
weight_kg            0
height_cm            0
gender              0
dtype: int64

```

Nombre de doublons (lignes strictement identiques) : 0

### 1.4.3 Liste complète des colonnes disponibles

```

[11]: print("\n--- Liste des colonnes du dataset ---")
      print(df.columns.tolist())

```

```

--- Liste des colonnes du dataset ---
['participant_id', 'date', 'age', 'gender', 'height_cm', 'weight_kg',
'activity_type', 'duration_minutes', 'intensity', 'calories_burned',
'avg_heart_rate', 'hours_sleep', 'stress_level', 'daily_steps',
'hydration_level', 'bmi', 'resting_heart_rate', 'blood_pressure_systolic',
'blood_pressure_diastolic', 'health_condition', 'smoking_status',
'fitness_level']

```

## 1.5 Statistiques descriptives et distribution des variables

### 1.5.1 Objectif

Obtenir une vue d'ensemble chiffrée des variables du dataset :

- résumer la distribution des variables numériques (moyenne, médiane, min, max, quartiles) ;
- analyser la répartition des variables catégorielles (genres, types d'activités, intensité, etc.) ;
- repérer les ordres de grandeur, les asymétries et les éventuelles valeurs extrêmes ;
- préparer l'identification de patterns intéressants pour la modélisation.

### 1.5.2 Interprétation attendue

À l'issue de cette étape, l'interprétation devra mettre en évidence :

- les valeurs centrales des variables clés (moyennes, médianes, étendue) pour caractériser le profil général des participants ;
- la dispersion et la variabilité des mesures (écarts-types, quartiles), permettant d'identifier des comportements hétérogènes ou des tendances majoritaires ;
- la présence éventuelle de valeurs extrêmes ou atypiques (poids très élevé, steps négatifs, calories inhabituellement hautes) qui pourront nécessiter un traitement spécifique ;
- la répartition des modalités des variables catégorielles (genre, types d'activités, intensité), révélant les comportements majoritaires ;

- les premières pistes d'analyse quant aux relations possibles entre activité physique, niveaux de stress, sommeil, hydratation et condition physique.

Cette interprétation servira de base pour orienter la suite de l'analyse, notamment l'identification des corrélations, la détection d'anomalies et la préparation des features pour la modélisation prédictive.

### 1.5.3 Statistiques descriptives globales

```
[13]: # Statistiques descriptives de base
stats = df.describe().T
display(stats)

# Vérification des bornes extrêmes sur les principales métriques du dataset
↳ FitLife
print(f"Âge moyen : {df['age'].mean():.1f} ans")
print(f"Poids moyen : {df['weight_kg'].mean():.1f} kg")
print(f"Calories moyennes brûlées : {df['calories_burned'].mean():.0f} kcal")
print(f"IMC moyen : {df['bmi'].mean():.1f}")
```

|                    | count    | mean        | std         | min     | 25%     |   |
|--------------------|----------|-------------|-------------|---------|---------|---|
| ↳ 50%              | 75%      | max         |             |         |         |   |
| participant_id     | 687701.0 | 1499.781828 | 865.997215  | 1.00    | 749.00  | ↳ |
| ↳1499.00           | 2249.00  | 3000.00     |             |         |         |   |
| age                | 687701.0 | 41.658602   | 13.581770   | 18.00   | 30.00   | ↳ |
| ↳42.00             | 53.00    | 64.00       |             |         |         |   |
| height_cm          | 687701.0 | 168.587699  | 9.140811    | 145.00  | 161.70  | ↳ |
| ↳168.20            | 175.30   | 198.50      |             |         |         |   |
| weight_kg          | 687701.0 | 94.921981   | 22.461801   | 45.30   | 78.20   | ↳ |
| ↳94.60             | 110.50   | 188.40      |             |         |         |   |
| duration_minutes   | 687701.0 | 70.011671   | 29.147251   | 20.00   | 45.00   | ↳ |
| ↳70.00             | 95.00    | 120.00      |             |         |         |   |
| calories_burned    | 687701.0 | 15.381302   | 9.985552    | 0.80    | 7.80    | ↳ |
| ↳13.00             | 20.70    | 92.00       |             |         |         |   |
| avg_heart_rate     | 687701.0 | 131.454165  | 17.814744   | 82.00   | 118.00  | ↳ |
| ↳130.00            | 144.00   | 206.00      |             |         |         |   |
| hours_sleep        | 687701.0 | 7.048799    | 0.972068    | 4.00    | 6.40    | ↳ |
| ↳ 7.00             | 7.70     | 10.00       |             |         |         |   |
| stress_level       | 687701.0 | 5.252389    | 2.770029    | 1.00    | 3.00    | ↳ |
| ↳ 5.00             | 8.00     | 10.00       |             |         |         |   |
| daily_steps        | 687701.0 | 8628.370918 | 2054.756608 | -419.00 | 7203.00 | ↳ |
| ↳8607.00           | 10027.00 | 17241.00    |             |         |         |   |
| hydration_level    | 687701.0 | 2.499427    | 0.579050    | 1.50    | 2.00    | ↳ |
| ↳ 2.50             | 3.00     | 3.50        |             |         |         |   |
| bmi                | 687701.0 | 22.733134   | 3.568383    | 14.20   | 20.10   | ↳ |
| ↳22.40             | 25.10    | 38.80       |             |         |         |   |
| resting_heart_rate | 687701.0 | 70.010547   | 5.074563    | 51.10   | 66.50   | ↳ |
| ↳70.00             | 73.50    | 87.10       |             |         |         |   |

|                          |          |            |           |       |        |   |
|--------------------------|----------|------------|-----------|-------|--------|---|
| blood_pressure_systolic  | 687701.0 | 120.001260 | 10.008917 | 78.00 | 113.10 | ⌵ |
| ↪ 120.20                 | 127.00   | 152.70     |           |       |        |   |
| blood_pressure_diastolic | 687701.0 | 80.188079  | 8.239418  | 53.70 | 74.60  | ⌵ |
| ↪ 80.10                  | 85.70    | 112.10     |           |       |        |   |
| fitness_level            | 687701.0 | 9.524900   | 5.502485  | 0.02  | 4.77   | ⌵ |
| ↪ 9.51                   | 14.23    | 21.93      |           |       |        |   |

Âge moyen : 41.7 ans  
 Poids moyen : 94.9 kg  
 Calories moyennes brûlées : 15 kcal  
 IMC moyen : 22.7

#### 1.5.4 Distribution des variables clés

```

[17]: num_vars = [
        "age",
        "weight_kg",
        "duration_minutes",
        "calories_burned",
        "daily_steps",
        "fitness_level"
      ]

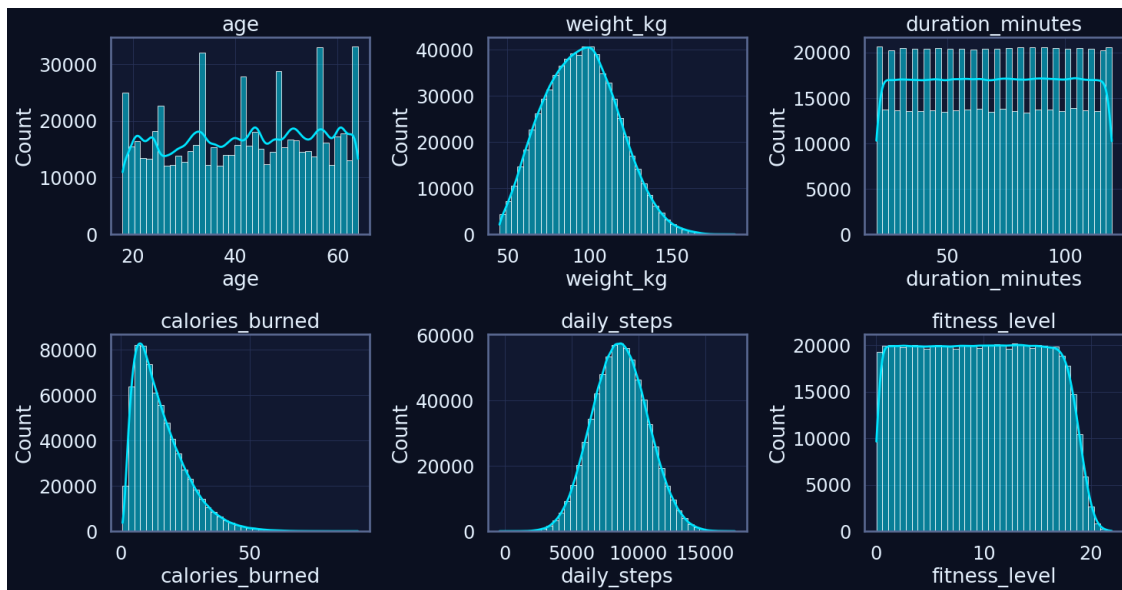
num_vars = [v for v in num_vars if v in df.columns]

n = len(num_vars)
n_cols = 3
n_rows = int(np.ceil(n / n_cols))

plt.figure(figsize=(5 * n_cols, 4 * n_rows))

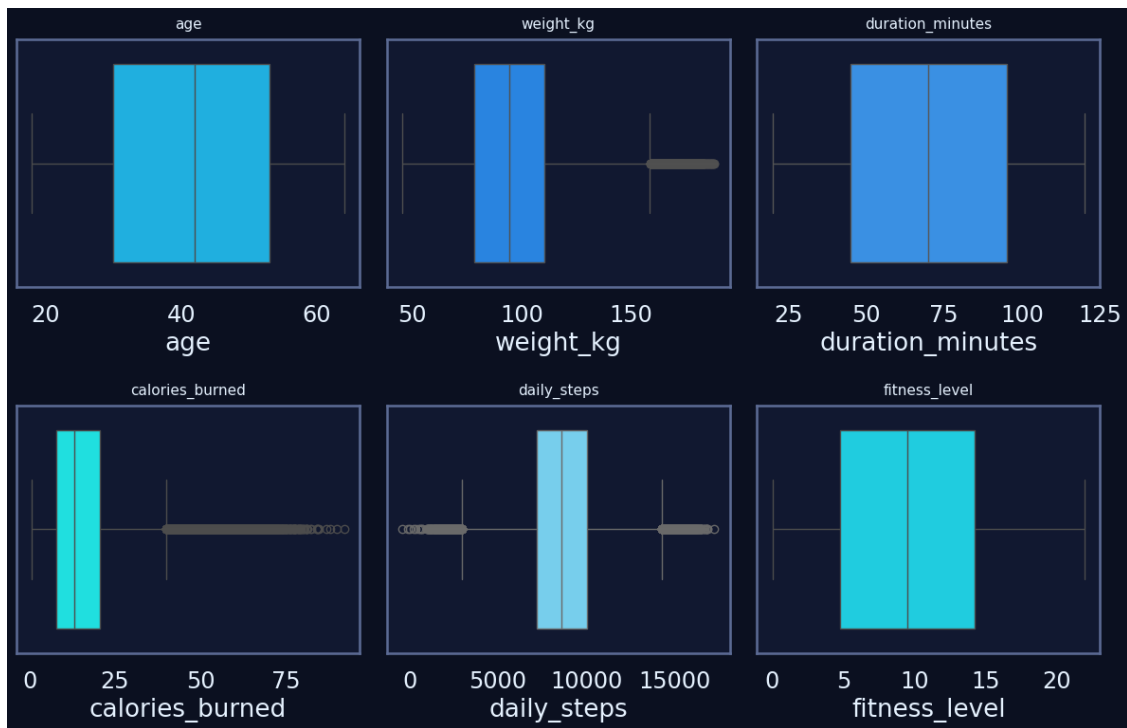
for i, col in enumerate(num_vars, 1):
    plt.subplot(n_rows, n_cols, i)
    sns.histplot(df[col], kde=True, bins=40)
    plt.title(col)
    plt.grid(True)

plt.tight_layout()
plt.show()
  
```



### 1.5.5 Boxplots pour repérer les outliers

```
[18]: plt.figure(figsize=(12, 8))
for i, col in enumerate(num_vars, 1):
    plt.subplot(2, 3, i)
    sns.boxplot(x=df[col], color=palette_trainme[i % len(palette_trainme)])
    plt.title(col, fontsize=11, pad=8)
plt.tight_layout()
plt.show()
```



### 1.5.6 Statistiques descriptives enrichies

```
[21]: # --- Colonnes numériques (on exclut les colonnes dérivées à magnitude extrême)
↳ ---
num_cols = df.select_dtypes(include="number").columns.tolist()
to_exclude = ["Burns Calories (per 30 min)_bc"] # valeurs ~1e19-1e20, non
↳ comparables
num_cols = [c for c in num_cols if c not in to_exclude]

print(f"{len(num_cols)} variables numériques analysées.")

desc = df[num_cols].describe().T
desc["missing_%"] = df[num_cols].isna().mean() * 100
desc["zeros_%"] = (df[num_cols] == 0).mean() * 100
desc["skew"] = df[num_cols].skew()
desc["kurt"] = df[num_cols].kurt()

display(desc.sort_values("std", ascending=False).head(12)) # top 12 par
↳ variance
```

16 variables numériques analysées.

|   | count | mean | std | min | 25% | 50% | 75% | max | missing_% |
|---|-------|------|-----|-----|-----|-----|-----|-----|-----------|
| ↳ |       |      |     |     |     |     |     |     |           |

|                          |          |             |             |         |         |   |
|--------------------------|----------|-------------|-------------|---------|---------|---|
| daily_steps              | 687701.0 | 8628.370918 | 2054.756608 | -419.00 | 7203.00 | ␣ |
| ↪8607.00                 | 10027.00 | 17241.00    | 0.0         |         |         |   |
| participant_id           | 687701.0 | 1499.781828 | 865.997215  | 1.00    | 749.00  | ␣ |
| ↪1499.00                 | 2249.00  | 3000.00     | 0.0         |         |         |   |
| duration_minutes         | 687701.0 | 70.011671   | 29.147251   | 20.00   | 45.00   | ␣ |
| ↪70.00                   | 95.00    | 120.00      | 0.0         |         |         |   |
| weight_kg                | 687701.0 | 94.921981   | 22.461801   | 45.30   | 78.20   | ␣ |
| ↪94.60                   | 110.50   | 188.40      | 0.0         |         |         |   |
| avg_heart_rate           | 687701.0 | 131.454165  | 17.814744   | 82.00   | 118.00  | ␣ |
| ↪130.00                  | 144.00   | 206.00      | 0.0         |         |         |   |
| age                      | 687701.0 | 41.658602   | 13.581770   | 18.00   | 30.00   | ␣ |
| ↪42.00                   | 53.00    | 64.00       | 0.0         |         |         |   |
| blood_pressure_systolic  | 687701.0 | 120.001260  | 10.008917   | 78.00   | 113.10  | ␣ |
| ↪120.20                  | 127.00   | 152.70      | 0.0         |         |         |   |
| calories_burned          | 687701.0 | 15.381302   | 9.985552    | 0.80    | 7.80    | ␣ |
| ↪13.00                   | 20.70    | 92.00       | 0.0         |         |         |   |
| height_cm                | 687701.0 | 168.587699  | 9.140811    | 145.00  | 161.70  | ␣ |
| ↪168.20                  | 175.30   | 198.50      | 0.0         |         |         |   |
| blood_pressure_diastolic | 687701.0 | 80.188079   | 8.239418    | 53.70   | 74.60   | ␣ |
| ↪80.10                   | 85.70    | 112.10      | 0.0         |         |         |   |
| fitness_level            | 687701.0 | 9.524900    | 5.502485    | 0.02    | 4.77    | ␣ |
| ↪9.51                    | 14.23    | 21.93       | 0.0         |         |         |   |
| resting_heart_rate       | 687701.0 | 70.010547   | 5.074563    | 51.10   | 66.50   | ␣ |
| ↪70.00                   | 73.50    | 87.10       | 0.0         |         |         |   |

|                          | zeros_% | skew      | kurt      |
|--------------------------|---------|-----------|-----------|
| daily_steps              | 0.0     | 0.062185  | -0.155186 |
| participant_id           | 0.0     | 0.000852  | -1.200757 |
| duration_minutes         | 0.0     | -0.001944 | -1.199195 |
| weight_kg                | 0.0     | 0.191474  | -0.378706 |
| avg_heart_rate           | 0.0     | 0.360952  | -0.234441 |
| age                      | 0.0     | -0.063646 | -1.193656 |
| blood_pressure_systolic  | 0.0     | -0.072306 | 0.008349  |
| calories_burned          | 0.0     | 1.235929  | 1.891749  |
| height_cm                | 0.0     | 0.179715  | -0.435383 |
| blood_pressure_diastolic | 0.0     | 0.132936  | 0.078316  |
| fitness_level            | 0.0     | 0.023257  | -1.164945 |
| resting_heart_rate       | 0.0     | 0.008333  | -0.016981 |

### 1.5.7 Détection rapide d'outliers (IQR)

```
[22]: iqr_report = {}

for col in num_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
```

```

IQR = Q3 - Q1
outliers = df[(df[col] < Q1 - 1.5 * IQR) | (df[col] > Q3 + 1.5 * IQR)]
iqr_report[col] = len(outliers)

iqr_df = pd.DataFrame.from_dict(iqr_report, orient="index",
    ↪columns=["outliers"])
iqr_df.sort_values("outliers", ascending=False).head(15)

```

```

[22]:
      outliers
calories_burned    17441
blood_pressure_diastolic    6107
hours_sleep        5897
bmi                 4147
resting_heart_rate    3738
blood_pressure_systolic    3602
daily_steps         3094
avg_heart_rate       1867
weight_kg            1706
height_cm            460
participant_id        0
age                   0
duration_minutes      0
stress_level          0
hydration_level       0

```

### 1.5.8 Aperçu corrélations (sous-ensemble lisible)

```

[24]: subset_cols = ["age", "weight_kg", "duration_minutes", "calories_burned",
    ↪"daily_steps", "fitness_level"]
corr = df[subset_cols].corr(numeric_only=True)
plt.figure(figsize=(7, 5))
sns.heatmap(
    corr,
    annot=True,
    fmt=".2f",
    square=True,
    cbar=True,
    annot_kws={"size": 8},      # taille des annotations
    linewidths=0.5,           # optionnel : lignes fines entre les cases
)

plt.title("Corrélations (sous-ensemble)", fontsize=12, pad=10)
plt.xticks(fontsize=9, rotation=45, ha="right")
plt.yticks(fontsize=9, rotation=0)
plt.tight_layout()
plt.show()

```

